2 DETERMINANTLAR NAZARIYASI

2.1. $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$ ikkinchi tartibli determinantni hisoblang:

Yechish. $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1 \cdot 4 - 2 \cdot 3 = -2.$ ▶

? Ikkinchi tartibli determinantni hisoblang.

2.2. $\begin{vmatrix} -7 & 6 \\ 5 & -4 \end{vmatrix}.$

2.3. $\begin{vmatrix} 10 & -5 \\ 9 & -8 \end{vmatrix}.$

2.4. $\begin{vmatrix} \sqrt{a}+\sqrt{b} & \sqrt{a}-\sqrt{b} \\ \sqrt{a}-\sqrt{b} & \sqrt{a}+\sqrt{b} \end{vmatrix}.$

2.5. $\begin{vmatrix} \sin 1^0 & \sin 89^0 \\ -\cos 1^0 & \cos 89^0 \end{vmatrix}.$

2.6. $\begin{vmatrix} \dfrac{x+y}{x} & \dfrac{2x}{x-y} \\ \dfrac{y-x}{x^2-y^2} & \dfrac{y-x}{x^2-y^2} \end{vmatrix}.$

2.7. $\begin{vmatrix} \sin^2\alpha & \cos^2\alpha \\ \sin^2\beta & \cos^2\beta \end{vmatrix}.$

2.8. $\begin{vmatrix} \sqrt{5}-a^{\frac{1}{2}} & a^{\frac{1}{2}} \\ -a^{\frac{1}{2}} & \sqrt{5}+a^{\frac{1}{2}} \end{vmatrix}.$

2.9. $\begin{vmatrix} \sin 60^0 & \cos 45^0 \\ \sin 45^0 & tg30^0 \end{vmatrix}.$

2.10. $\begin{vmatrix} tga & -1 \\ 4 & ctga \end{vmatrix}.$

2.11. $\begin{vmatrix} 1,(3) & 2,25 \\ 23/3 & 6 \end{vmatrix}.$

2.12. $\begin{vmatrix} \dfrac{a-1}{2\sqrt{a}} & \dfrac{a+\sqrt{a}}{\sqrt{a}-1} \\ \dfrac{a\sqrt{a}-\sqrt{a}}{2a} & \dfrac{a-\sqrt{a}}{\sqrt{a}+1} \end{vmatrix}.$

? Tenglamani yeching.

2.13. $\begin{vmatrix} 2x+1 & 3 \\ x+5 & 2 \end{vmatrix} = 0.$

2.14. $\begin{vmatrix} x+3 & x-1 \\ 7-x & x-1 \end{vmatrix} = 0.$

2.15. $\begin{vmatrix} 2x-1 & x+1 \\ x+2 & x-1 \end{vmatrix} = -6.$

2.16. $\begin{vmatrix} x-2 & y+3 \\ -y-3 & x-2 \end{vmatrix} = 0.$

2.17. $\begin{vmatrix} \sin 2x & \sin x \\ \cos x & \cos 2x \end{vmatrix} = 0.$

2.1. Uchinchi tartibli determinantni hisoblang: $\begin{vmatrix} 3 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 4 & 2 \end{vmatrix}$.

Yechish. Determinantni birinchi satr elementlari bo‘yicha yoyib hisoblaymiz: $\begin{vmatrix} 3 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 4 & 2 \end{vmatrix} = 3 \cdot \begin{vmatrix} 5 & 3 \\ 4 & 2 \end{vmatrix} - 2 \cdot \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} + 1 \cdot \begin{vmatrix} 2 & 5 \\ 3 & 4 \end{vmatrix} =$

$$= 3 \cdot (5 \cdot 2 - 3 \cdot 4) - 2 \cdot (2 \cdot 2 - 3 \cdot 3) + 1 \cdot (2 \cdot 4 - 5 \cdot 3) =$$

$$= 3 \cdot (-2) - 2 \cdot (-5) + 1 \cdot (-7) = -3.$$

Uchinchi tartibli determinantlarni ixtiyoriy satr (ustun) elementlari bo‘yicha yoyib hisoblang.

2.2. $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$.

2.3. $\begin{vmatrix} 2 & 1 & 3 \\ 5 & 3 & 2 \\ 1 & 4 & 3 \end{vmatrix}$.

2.4. $\begin{vmatrix} 3 & 2 & -1 \\ -2 & 2 & 3 \\ 4 & 2 & -3 \end{vmatrix}$.

2.5. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{vmatrix}$.

2.6. $\begin{vmatrix} 2 & 1 & 3 \\ 4 & 5 & 9 \\ 16 & 25 & 81 \end{vmatrix}$.

2.7. Uchinchi tartibli determinantni uchburchak qoidasidan foydalanib hisoblang: $\begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & -6 \\ 7 & 8 & 9 \end{vmatrix}$.

Yechish. $\begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & -6 \\ 7 & 8 & 9 \end{vmatrix} = 1\cdot 5\cdot 9 + 2\cdot(-6)\cdot 7 + (-4)\cdot 3\cdot 8 -$

$-3\cdot 5\cdot 7 - (-4)\cdot 2\cdot 9 - 1\cdot(-6)\cdot 7 = 45 - 84 - 96 - 105 + 72 + 42 = 126.$ ▶

? Uchburchak qoidasidan foydalanib determinantlarni hisoblang.

2.8. $\begin{vmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{vmatrix}$.

2.9. $\begin{vmatrix} 0 & x & 0 \\ y & 0 & 0 \\ 0 & 0 & z \end{vmatrix}$.

2.10. $\begin{vmatrix} 0 & 1 & 0 \\ 2 & 3 & 4 \\ 0 & 5 & 0 \end{vmatrix}$.

? Uchinchi tartibli determinantlarni qulay usulda hisoblang.

2.11. $\begin{vmatrix} 2 & 3 & 4 \\ 5 & -2 & 1 \\ 1 & 2 & 3 \end{vmatrix}$.

2.12. $\begin{vmatrix} a & 1 & a \\ -1 & a & 1 \\ a & -1 & a \end{vmatrix}$.

2.13. $\begin{vmatrix} 1 & 2 & 3 \\ 8 & 1 & 4 \\ 2 & 1 & 1 \end{vmatrix}$.

2.14. $\begin{vmatrix} 5 & 3 & 2 \\ -1 & 2 & 4 \\ 7 & 3 & 6 \end{vmatrix}$.

2.15. $\begin{vmatrix} 3 & -1 & -2 \\ 1 & 2 & 5 \\ -4 & 1 & 6 \end{vmatrix}$.

2.16. $\begin{vmatrix} 1 & 2 & -1 \\ 3 & 7 & 2 \\ 2 & 3 & -7 \end{vmatrix}$.

2.17. $\begin{vmatrix} a & -a & a \\ a & a & -a \\ a & -a & -a \end{vmatrix}$.

2.18. $\begin{vmatrix} a+x & x & x \\ x & b+x & x \\ x & x & c+x \end{vmatrix}$.

2.19. $\begin{vmatrix} \cos\alpha & \sin\alpha\cos\beta & \sin\alpha\sin\beta \\ -\sin\alpha & \cos\alpha\cos\beta & \cos\alpha\sin\beta \\ 0 & -\sin\beta & \cos\beta \end{vmatrix}$.

2.20. $\begin{vmatrix} x^2 & x & 1 \\ y^2 & y & 1 \\ z^2 & z & 1 \end{vmatrix}.$

2.21. $\begin{vmatrix} m+a & m-a & a \\ n+a & 2n-a & a \\ a & -a & a \end{vmatrix}.$

2.22. $\begin{vmatrix} ax & a^2+x^2 & 1 \\ ay & a^2+y^2 & 1 \\ az & a^2+z^2 & 1 \end{vmatrix}.$

2.23. $\begin{vmatrix} \sin 3\alpha & \cos 3\alpha & 1 \\ \sin 2\alpha & \cos 2\alpha & 1 \\ \sin\alpha & \cos\alpha & 1 \end{vmatrix}.$

2.24. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}.$

2.25. $\begin{vmatrix} a & x & x \\ x & b & x \\ x & x & c \end{vmatrix}.$

? Tenglama va tengsizlikni yeching.

2.26. $\begin{vmatrix} 2 & 0 & 3 \\ -1 & 7 & x-3 \\ 5 & -3 & 6 \end{vmatrix} = 0.$

2.27. $\begin{vmatrix} -1 & 3 & -2 \\ 2-3x & 0 & 5 \\ 3 & 2 & 1 \end{vmatrix} \geq 0.$

2.28. $\begin{vmatrix} -1 & 0 & 2x+3 \\ 3-x & 1 & 1 \\ 2x+1 & -1 & 2 \end{vmatrix} = 0.$

2.29. $\begin{vmatrix} 6 & 3 & x-1 \\ 2x & 1 & 0 \\ 4 & x+2 & 2 \end{vmatrix} = 0.$

2.30. Determinantning xossalaridan foydalanib tenglikni isbotlang:

$$\begin{vmatrix} a_1 & b_1 & c_1+a_1x+b_1y \\ a_2 & b_2 & c_2+a_2x+b_2y \\ a_3 & b_3 & c_3+a_3x+b_3y \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$$

Isbot. Chap determinantning uchunchi ustunini uchta ustun yig‘indisi ko‘rinishida ifodalash mumkin, bu determinantni uchta determinant yig‘indisi ko‘rinishida ifodalaymiz:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} a_1 & b_1 & a_1x \\ a_2 & b_2 & a_2x \\ a_3 & b_3 & a_3x \end{vmatrix} + \begin{vmatrix} a_1 & b_1 & b_1y \\ a_2 & b_2 & b_2y \\ a_3 & b_3 & b_3y \end{vmatrix}.$$

Ikkinchi determinantning uchunchi ustuni birinchi ustuniga proporsional, uchunchi determinantning uchunchi ustuni ikkinchi ustuniga proporsional. Shuning uchun ikkinchi va uchinchi determinantlar nolga teng. ▶

?

Tenglikni isbotlang.

2.31. $\begin{vmatrix} a_1+b_1x & a_1-b_1x & c_1 \\ a_2+b_2x & a_2-b_2x & c_2 \\ a_3+b_3x & a_3-b_3x & c_3 \end{vmatrix} = -2x \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$

2.32. $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (b-a)(c-a)(c-b).$

2.33. $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (b-a)(c-a)(c-b).$

2.34. $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}.$

2.35. $\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a+b+c)\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}.$

2.36. $\begin{vmatrix} (a+b)^2 & c^2 & c^2 \\ a^2 & (b+c)^2 & a^2 \\ b^2 & b^2 & (c+a)^2 \end{vmatrix} = 2bc(a+b+c)^3.$

2.37. To‘rtinchi tartibli determinantni hisoblang:

$$\Delta = \begin{vmatrix} a & 0 & 3 & 5 \\ 0 & 0 & b & 2 \\ 1 & c & 2 & 3 \\ 0 & 0 & 0 & d \end{vmatrix}.$$

Determinantni to‘rtinchi satr elementlari bo‘yicha yoyib hisoblaymiz:

Yechish. $\Delta = (+d)\cdot\begin{vmatrix} a & 0 & 3 \\ 0 & 0 & b \\ 1 & c & 2 \end{vmatrix} = \begin{bmatrix} \text{det}\,er\,\text{min}\,antni \\ 2-satr\,bo'yicha\,yoyamiz \end{bmatrix} =$

$= d\cdot(-b)\cdot\begin{vmatrix} a & 0 \\ 1 & c \end{vmatrix} = -d\cdot b\cdot a\cdot c.$ ▶

?

To‘rtinchi tartibli determinantni hisoblang.

2.38. $\begin{vmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 3 \\ 3 & 0 & 1 & 2 \\ 2 & 3 & 0 & 1 \end{vmatrix}.$

2.39. $\begin{vmatrix} 1 & 2 & 0 & -3 \\ 3 & 1 & 0 & 4 \\ 1 & 5 & -1 & 7 \\ -2 & 1 & 0 & 1 \end{vmatrix}.$

2.40. $\begin{vmatrix} 1 & 2 & 3 & 4 \\ -9 & -9 & -9 & -9 \\ 4 & 3 & 2 & 1 \\ 1 & 0 & 1 & 0 \end{vmatrix}.$

2.41. $\begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 5 & 9 \\ 0 & 0 & 3 & 7 \\ -2 & -4 & -6 & 1 \end{vmatrix}.$

2.42. $\begin{vmatrix} 1 & 2 & -3 & 1 \\ 3 & 0 & 1 & -1 \\ 2 & 0 & 4 & 1 \\ 5 & 1 & 2 & 1 \end{vmatrix}.$

2.43. $\begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{vmatrix}.$

2.44. $\begin{vmatrix} 1 & -2 & 3 & 4 \\ 2 & 1 & -4 & 3 \\ 3 & -4 & -1 & -2 \\ 4 & 3 & 2 & -1 \end{vmatrix}.$

2.45. $\begin{vmatrix} -1 & -1 & -1 & -1 \\ -1 & -2 & -4 & -8 \\ -1 & -3 & -9 & -27 \\ -1 & -4 & -16 & -64 \end{vmatrix}.$

2.46. $\begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1-a & 1 & 1 \\ 1 & 1 & 1+b & 1 \\ 1 & 1 & 1 & 1-b \end{vmatrix}.$

2.47. $\begin{vmatrix} 3 & -1 & 2 & -1 & 1 \\ 5 & 1 & -2 & 1 & 2 \\ 9 & -1 & 1 & 3 & 4 \\ 3 & 0 & 6 & -1 & 3 \\ 5 & 2 & 3 & -2 & 1 \end{vmatrix}.$

2.48. $\begin{vmatrix} 0 & 6 & 3 & 5 & 1 \\ -3 & 2 & 4 & 1 & 0 \\ 5 & 1 & 4 & 3 & 2 \\ -3 & 8 & 7 & 6 & 1 \\ 1 & 0 & 3 & 4 & 0 \end{vmatrix}.$